

Confocal Membrane Transport Atlas

Membrane flux measurements in Arabidopsis thaliana

Abstract

Membrane flux was estimated from background-corrected confocal sequences while tracking a fixed region of interest. Spectral bleed-through was measured independently so that the atlas would not assign transport to detector cross-talk.

1. Imaging

Seedling epidermal cells of Arabidopsis thaliana were imaged with marker MTR-7. Excitation was fixed at $\lambda_{ex}=488$ nm and detector gain remained unchanged.

1.1 Correction

Background correction used a cell-free region adjacent to ROI_A3. No intensity normalization was applied after subtraction.

1.2 Flux estimate

Membrane flux was derived only during intervals with stable focus. Na⁺/H⁺ labels identify the transport condition and are not algebraic operators.

Scope

The atlas reports spatial association and does not establish a molecular transport mechanism.

$$F_{corr} = F_{raw} - B_{\lambda}$$

2. Confocal panels and spectral map

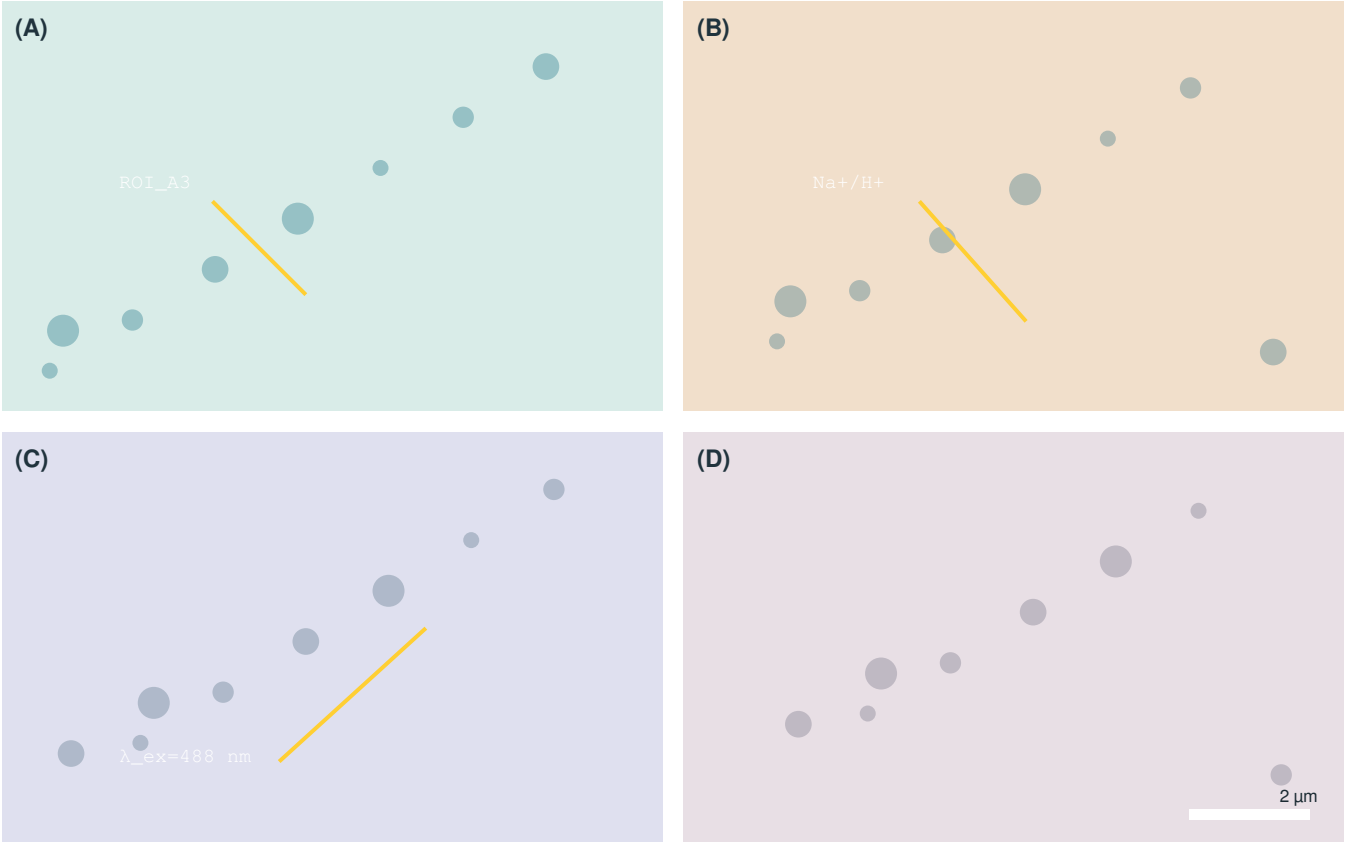


Figure 2. Panels (A)-(D) show MTR-7 signal, ROI_A3, Na+/H+ condition, and the spectral heatmap. Scale bars are 2 μm and 500 nm.

Panel	Cells	F_raw	B_λ	F_corr
(A)	18	1.000	0.083	0.917
(B)	16	0.842	0.071	0.771
(C)	17	1.126	0.098	1.028

Table 1. Background correction by panel. Values retain three decimal places.

3. Results

Corrected signal increased within ROI_A3 after the Na⁺/H⁺ condition, but the effect was heterogeneous across cells. Spectral bleed-through remained below 9.8 % of F_{raw}.

3.1 Limitations

Focus drift and membrane curvature constrain the apparent membrane flux. The compact atlas is descriptive and does not distinguish transport direction.

Footnote

Footnote 1. ROI_A3 was defined before treatment and was not redrawn after inspection.

References

- [1] Tsai L. Background correction in confocal time series. *Imaging Methods* 15:30-39.
- [2] Moreau J. Spectral bleed-through controls for membrane markers. *Cell Optics* 8:111-120.
- [3] Singh P. Reporting regions of interest without selection bias. *Microscopy Notes* 4:51-57.